# 7. Mathematics for Decision Making

## 7.1 Introduction to Linear Programming

The idea in linear programming (LP) is to find the maximum (or minimum) value of a linear function subject to constraints. Consider the following scenario.

A gas processing plant receives a fixed amount of raw gas (77 $m^3$) and can use it to produce two grades of gas, regular or premium. The profit for producing one tonne of regular gas is £150 while the profit for one tonne of premium gas is £175. The goal is to decide how much gas of each type to produce to maximize the profit subject to certain constraints.

Letting $x$ represent the amount (in tonnes) of regular gas and $y$ the amount of premium gas, the overall profit is:

$$Z = 150x + 175y.$$

This is called the **objective function** and $x$ and $y$ are **decision variables**.

If 11 $m^3$ of raw gas is needed for one tonne of either regular or premium gas, the solution would be easy. Just make as much premium gas as possible, which in this case is 7 tonnes giving a profit of £175 x 7 = £1225.

**Material Constraint**

However, suppose that while 11 $m^3$ of raw gas is needed for one tonne of premium gas only 7 $m^3$ of raw gas is needed for one tonne of regular gas. This gives rise to the following constraint

$$7x + 11y \leq 77.$$

If this were the only constraint, the solution would still be fairly straightforward. Instead of producing 7 tonnes of premium gas, 11 tonnes of raw gas could be produced giving a profit of £150 x 11 = £1650. (There's also no advantage in producing some premium gas, e.g. if one tonne of premium were to be produced, that would leave 66 $m^3$ of raw gas, which could be used to make 66/7 $\approx$ 9.43 tonnes of regular, but that gives a profit of just £150 x 9.43 + 175 x 1 = £1589.)

Suppose, however, that there are further constraints.

**Time Constraint**

Only one type of gas can be produced at a time - regular requires 10 hours per tonne and premium 8 hours - and the total time available is 80 hours. This constraint can be expressed as follows

$$10x + 8y \leq 80.$$

**Storage Constraint** A maximum of 9 tonnes of regular gas and 6 tonnes of premium gas can be stored. These constraints can be expressed as follows

$$x \leq 9,$$
$$y \leq 6.$$

It's now not so obvious how to find the optimal solution, so we need to use LP.

## 7.2 Formulating a Linear Programming Problem

Here are a number of steps for formulating a LP problem:

1. Identify all relevant constraints and the objective.
2. Identify decision variables and whether any could have negative values.
3. Express the objective function in terms of the decision variables.
4. Express the constraints as inequalities.

The gas processing problem can be expressed as a LP problem as follows:

| | | |
|---|---|---|
| Maximize: | $Z = 150x + 175y$ | |
| Subject to: | $7x + 11y \leq 77$ | (material constraint) |
| | $10x + 8y \leq 80$ | (time constraint) |
| | $x \leq 9$, $y \leq 6$ | (storage constraints) |
| | $x \geq 0$, $y \geq 0$ | (non-negativity constraints) |

The non-negativity constraints weren't specified in section 7.1, but they just represent the obvious constraint that we can't have a solution with a negative amount of regular or premium gas.

**Exercise 7.1**

A truck rental company rents two types of trucks. Type A has 100 cubic feet of refigerated available space and 200 cubic feet of non-refigerated space. Type B has 150 cubic feet of each type of space. A transportation company needs to ship at least 4500 cubic feet of refigerated product and at least 6000 cubic feet of non-refigerated product. The problem they want to solve is to determine how many trucks of each type to rent if Type A costs £3 per mile and Type B costs £4 per mile.

Formulate this as a LP problem making sure to identify decision variables, constraints and the objective function.

*Solution*

1. There are constraints in terms of a) the refigerated space, b) the non-refigerated space. The objective is not stated directly, but it can be assumed that it is to minimize the cost per mile.
2. Decision variables. Let $x$ be the number of type A trucks and $y$ the number of type B trucks. Clearly, neither could be negative so this gives non-negativity constraints.
3. Objective function is $Z = 3x + 4y$.
4. Inequalities for constraints - see below.

| | | |
|---|---|---|
| Minimize: | $Z = 3x + 4y$ | |
| Subject to: | $100x + 150y \geq 4500$ | (refigerated constraint) |
| | $200x + 150y \geq 6000$ | (non-refigerated constraint) |
| | $x \geq 0$, $y \geq 0$ | (non-negativity constraints) |

**Standard Maximization Problem (SMP)**

So far we have only considered two decision variables, $x$ and $y$, but consider the following LP problem with decision variables $x_1, x_2, \ldots, x_n$

| | |
|---|---|
| Maximize: | $Z = c_1x_1 + c_2x_2 + \ldots c_nx_n$ |
| Subject to: | $a_{11}x_1 + a_{12}x_2 + \ldots a_{1n}x_n \leq b_1$ |
| | $a_{21}x_1 + a_{22}x_2 + \ldots a_{2n}x_n \leq b_2$ |
| | $\vdots$ |
| | $a_{k1}x_1 + a_{k2}x_2 + \ldots a_{kn}x_n \leq b_k$ |
| | $x_1, x_2, \ldots, x_n \geq 0$ |

The values $c_1, c_2, \ldots, c_n$ are objective (or cost) coefficients, the $a_{11}, \ldots a_{kn}$ are constraint coefficients and the $b_1, \ldots b_k$ are resource constants.

For a LP problem to be a SMP it must maximize the objective function, the resource constraints must be of the form above (i.e. $\ldots \leq b_j$), the resource constants $b_1, \ldots b_k$ must be non-negative and the decision variables must be non-negative.

Every LP can actually be converted into SMP form. See the textbook C. Morris and R. Stark, *Finite Mathematics*, Wiley (2016) for details.

**Exercise 7.2**

Determine whether a) the formulation of the gas processing LP problem and b) the formulation of the truck rental problem in the solution to Exercise 7.1 are in SMP form.

*Solution*

a) The gas processing problem is in SMP form since it maximizes the objective function, has the right kinds of contraints with the resource constants all positive (77, 80, 9, 6) and non-negative decision variables.

b) The LP problem in Exercise 7.1 is not in SMP form since it minimizes the objective function and also has the wrong kinds of contraints (e.g. . . . $\geq$ 4500).

## 7.3 Linear Programming - Graphical Solution

Let's now see how to solve a LP problem graphically. We will consider the gas processing LP problem that we formulated earlier.

First of all, plot the material constraint represented by the inequality

$$7x + 11y \leq 77.$$

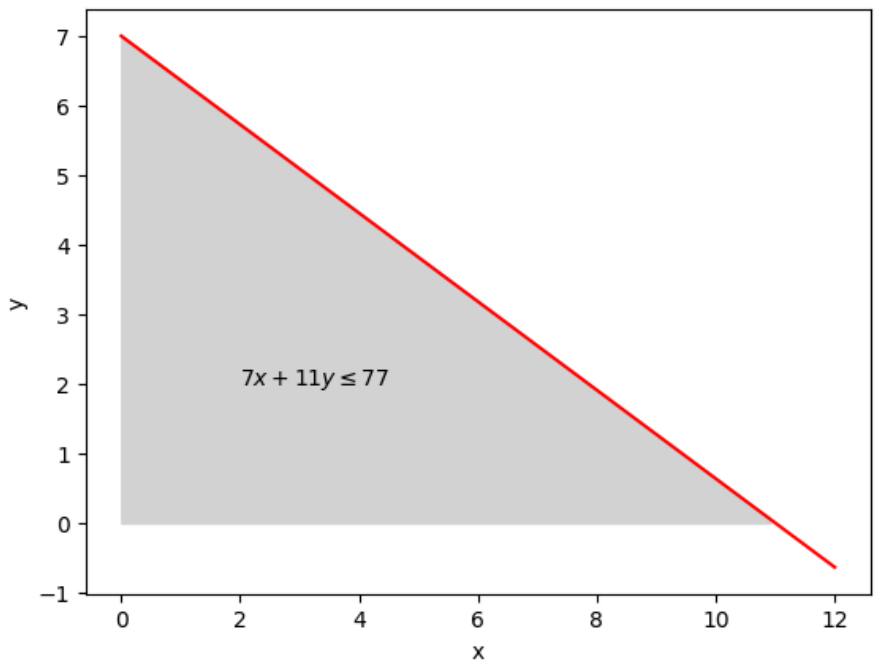
To do this, we can plot the line $7x + 11y = 77$ and then shade the side of it that satisfies the inequality.

Rearranging the equation to express it in the form $y = mx + c$ gives

$$y = -\frac{7}{11}x + 7.$$

The shaded area in the figure below shows the region of values of $x$ and $y$ where this constraint is satisfied. (Note that we have also assumed that $x$ and $y$ cannot be negative.)

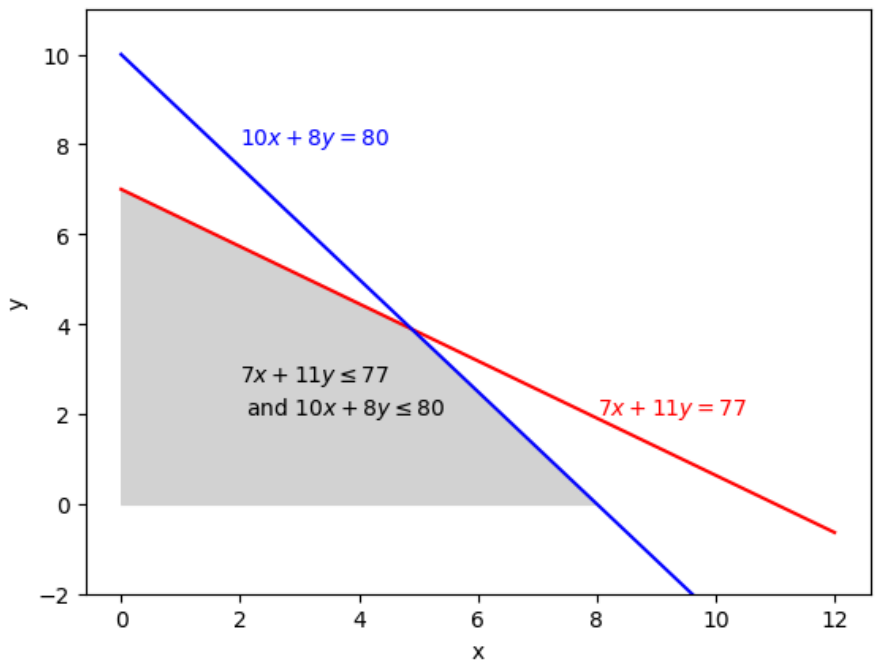
In [ ]:
```
import numpy as np
import matplotlib.pyplot as plt

# Inclusion of material constraint
x = np.linspace(0,12,100)
y = -7/11 * x + 7
plt.plot(x, y,color='red')

# Region satisfying constraint
plt.fill_between(x,y,where=y>=0,color='lightgrey')

plt.xlabel('x')
plt.ylabel('y')
plt.text(2, 2, '$7x + 11y \leq 77$')

plt.show()
```




The other constraints can also be added in the same way. The figure below shows the inclusion of the time constraint

$$10x + 8y \leq 80.$$

In [ ]:
```
# Inclusion of material constraint
x = np.linspace(0,12,100)
y = -7/11 * x + 7
plt.plot(x, y,color='red')

# Inclusion of time constraint
y2 = -5/4 * x + 10
plt.plot(x, y2, color='blue')

# Region satisfying all constraints so far
y_min = np.minimum(y,y2)
plt.fill_between(x,y_min,where=y_min>=0,color='lightgrey')

plt.ylim(-2, 11)
plt.xlabel('x')
plt.ylabel('y')

plt.text(2, 8, '$10x + 8y = 80$', color='blue')
plt.text(8, 2, '$7x + 11y = 77$', color='red')
plt.text(2, 2, '$7x + 11y \leq 77$\n and $10x + 8y \leq 80$')

plt.show()
```




Finally, adding in the storage constraints

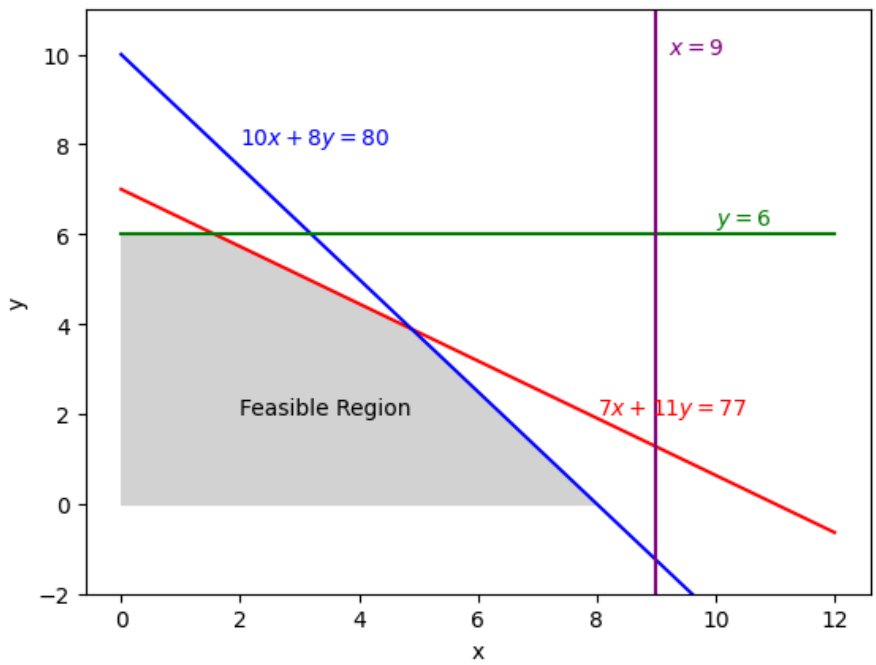
$$x \leq 9,$$

$$y \leq 6.$$

gives the figure below. The shaded region gives the **Feasible Region** - the region of possible solutions to the problem.

(Note that the storage constraint for $x$, the amount of regular gas, does not constrain the feasible region beyond the other constraints that have already been included.)

In [ ]:
```
# Inclusion of material constraint
x = np.linspace(0,12,100)
y = -7/11 * x + 7
plt.plot(x, y,color='red')

# Inclusion of time constraint
y2 = -5/4 * x + 10
plt.plot(x, y2, color='blue')

# Inclusion of storage constraint for y (premium gas)
y3 = 6 * np.ones_like(x)
plt.plot(x, y3, color='green')

# Inclusion of storage constraint for x (regular gas)
y4 = np.array([-2,11])
x4 = 9 * np.ones_like(y4)
plt.plot(x4, y4, color='purple')

# Region satisfying all constraints
y_min = np.minimum(y,np.minimum(y2,y3))
plt.fill_between(x,y_min,where=y_min>=0,color='lightgrey')

plt.ylim(-2, 11)
plt.xlabel('x')
plt.ylabel('y')

plt.text(2, 8, '$10x + 8y = 80$', color='blue')
plt.text(8, 2, '$7x + 11y = 77$', color='red')
plt.text(10, 6.2, '$y = 6$', color='green')
plt.text(9.2, 10, '$x = 9$', color='purple')
plt.text(2, 2, 'Feasible Region')

plt.show()
```




Having identified the feasible region of possible solutions, we still need to find the optimal solution that maximizes profit.

To do this, we can plot multiple lines representing specified values of profit $Z$. Recall the equation for the profit

$$Z = 150x + 175y,$$

or to put it in the form $y = mx + c$,

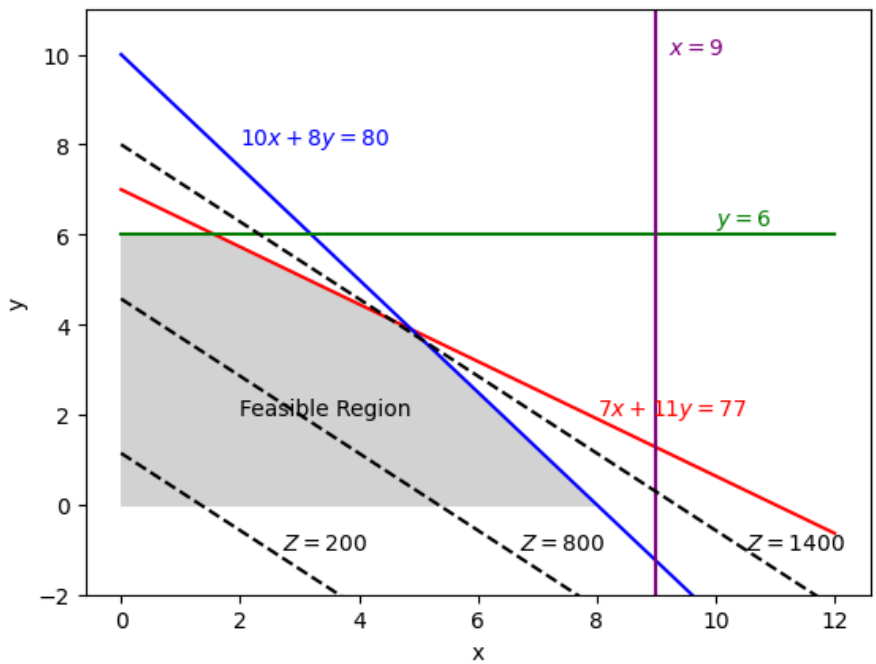
$$y = -\frac{150}{175} + \frac{Z}{175}.$$

By putting in various values of $Z$ and plotting them we get the dashed lines in the figure below. Consider the line for $Z = 800$. Every point along this line in the shaded area (the feasible region) represents values of $x$ and $y$ that give a profit of £800. Clearly, it is possible to get higher profit, but the maximum value of $Z$ seems to be about £1400 since if we go much higher than that it won't overlap with the feasible region.

The solution occurs at the corner of the feasible region. In this case it is where the blue and red lines cross. (In other cases, it might be at a different corner of the feasible region.) Graphically, this occurs close to $x = 5$ and $y = 4$, giving a profit of about £1400, but we can get a more precise solution.

In [ ]:
```
# Inclusion of material constraint
x = np.linspace(0,12,100)
y = -7/11 * x + 7
plt.plot(x, y,color='red')

# Inclusion of time constraint
y2 = -5/4 * x + 10
plt.plot(x, y2, color='blue')

# Inclusion of storage constraint for y (premium gas)
y3 = 6 * np.ones_like(x)
plt.plot(x, y3, color='green')

# Inclusion of storage constraint for x (regular gas)
y4 = np.array([-2,11])
x4 = 9 * np.ones_like(y4)
plt.plot(x4, y4, color='purple')

# Region satisfying all constraints
y_min = np.minimum(y,np.minimum(y2,y3))
plt.fill_between(x,y_min,where=y_min>=0,color='lightgrey')

# Profit lines
Z = 200
y5 = -150/175 * x + Z/175
plt.plot(x, y5, color='black', linestyle='dashed')
```

```
Z = 800
y6 = -150/175 * x + Z/175
plt.plot(x, y6, color='black', linestyle='dashed')
Z = 1400
y6 = -150/175 * x + Z/175
plt.plot(x, y6, color='black', linestyle='dashed')

plt.ylim(-2, 11)
plt.xlabel('x')
plt.ylabel('y')

plt.text(2, 8, '$10x + 8y = 80$', color='blue')
plt.text(8, 2, '$7x + 11y = 77$', color='red')
plt.text(10, 6.2, '$y = 6$', color='green')
plt.text(9.2, 10, '$x = 9$', color='purple')
plt.text(2.7, -1, '$Z =200$')
plt.text(6.7, -1, '$Z =800$')
plt.text(10.5, -1, '$Z =1400$')
plt.text(2, 2, 'Feasible Region')

plt.show()
```




To find the point where the red and blue lines cross, we need to solve the equations for these lines simultaneously, i.e. we need to solve the set of equations

$$7x + 11y = 77$$

$$10x + 8y = 80.$$

We can use the approach from section 5.4 of the lecture notes as is done in the cell below. This gives the precise values of $x$ and $y$ which in turn enable us to find the corresponing proft from the equation

$$Z = 150x + 175y.$$

In [ ]:
```
A = np.array([[7,11],[10,8]])
rhs = np.array([77,80]).T

x,y = np.linalg.solve(A,rhs)
print(x,y)

# Profit for solution
Z =150 * x + 175 * y
print('Z: ', Z)

print('\nRegular gas: {:5.3f} tonnes'.format(x))
print('Premium gas: {:5.3f} tonnes'.format(y))
print('Profit: £{:4.0f}'.format(Z))
```

```
4.88888888888889 3.888888888888888
Z:  1413.888888888889

Regular gas: 4.889 tonnes
Premium gas: 3.889 tonnes
Profit: £1414
```

Hence, the optimal solution is to produce 4.889 tonnes of regular gas and 3.889 tonnes of premium gas, resulting in a profit of £1414.

**Exercise 7.3**

Sketch a graphical solution to the truck rental problem in Exercise 7.1.

# 7.4 Algorithmic Solutions to Linear Programming Problems

In section 7.3, we saw that the gas processing LP problem had a solution at one of the corners of the *feasbile region*. It turns out that this is a general feature of LP problems.

If the optimal value of the objective function in a LP problem exists, then that value must occur at one (or more) of the corner points of the feasible region.

This result is used in the popular Simplex Method for solving LP problems. A detailed discuss of this approach is beyond the scope here, but the way it works is by moving from one corner of the feasible region to an adjacent corner increasing the value of the objective function each time until the optimal solution is obtained. See the textbook C. Morris and R. Stark, *Finite Mathematics*, Wiley (2016) for details.

Here we will use the Python library PuLP which can be used for solving a range of optimization problems.

Note: you will need to ensure that you have PuLP installed - see Installing PuLP at Home

Let's see how to use this package to solve the gas procecssing problem. Here are some key steps

- import pulp library
- create problem object and specify whether to maximize or minimize

`prob = pl.LpProblem('Gas_Problem', pl.LpMaximize)`

- create decision variables and specify as non-negative, e.g.

`x = pl.LpVariable("x", lowBound = 0)`

- specify constraints, e.g.

`prob += 7 * x + 11 * y <= 77`

Note that they are added to the problem object.

- call the solver (the default solver is called CBC (Coin-or Branch and Cut) which makes use of the Simplex method)

`prob.solve()`

- display solution

In [ ]:
```
# Use following line to install pulp library
#!pip install pulp

# import the library pulp
import pulp as pl

# Create LP problem object to contain problem information
prob = pl.LpProblem('Gas_Problem', pl.LpMaximize)

# Create decision variables
x = pl.LpVariable("x", lowBound = 0)   # Create a variable x >= 0
y = pl.LpVariable("y", lowBound = 0)   # Create a variable y >= 0

# Add objective function to problem
```

```python
prob += 150 * x + 175 * y

# Add constraints to problem
prob += 7 * x + 11 * y <= 77
prob += 10 * x + 8 * y <= 80
prob += x <= 9
prob += y <= 6

# Display problem
print(prob)

# Solve the problem using default solver
prob.solve()

# Check status of solution - is it optimal?
print('Status :', pl.LpStatus[prob.status])

# Display the solution
print('\nSolution :')
print(pl.value(x), pl.value(y), pl.value(prob.objective))
print('\nRegular gas: {:5.3f} tonnes'.format(pl.value(x)))
print('Premium gas: {:5.3f} tonnes'.format(pl.value(y)))
print('Profit: £{:4.0f}'.format(pl.value(prob.objective)))
```

```
Gas_Problem:
MAXIMIZE
150*x + 175*y + 0
SUBJECT TO
_C1: 7 x + 11 y <= 77

_C2: 10 x + 8 y <= 80

_C3: x <= 9

_C4: y <= 6

VARIABLES
x Continuous
y Continuous

Status : Optimal

Solution :
4.8888889 3.8888889 1413.8888925

Regular gas: 4.889 tonnes
Premium gas: 3.889 tonnes
Profit: £1414
```

**Exercise 7.4**

Suppose that there is an opportunity either to increase the amount of raw gas by $5m^3$ or to increase the time available by 5 hours without changing the amount of profit per tonne. Which option would permit the most profitable solution?

*Solution*

Increasing the amount of raw gas to 82 gives the following result:

Regular gas: 4.148 tonnes

Premium gas: 4.815 tonnes

Profit: £1465

Increasing the amount of raw gas to 85 gives the following result:

Regular gas: 5.907 tonnes

Premium gas: 3.241 tonnes

Profit: £1453

So it is more profitable to increase the volume of raw gas.